

COURSE CODE: CSA16

COURSE NAME: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3, BL4, BL5)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

EXPERIMENTS:

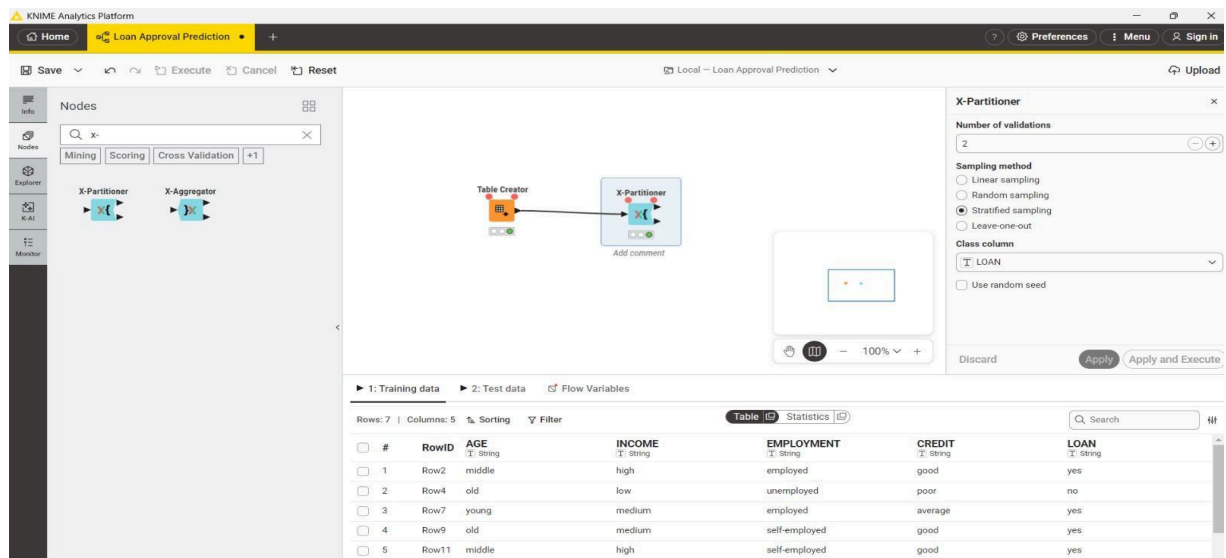
3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**.
(10 Marks)

Consider the Data Set

young,high,employed,good,yes
young,high,self-employed,average,yes
middle,high,employed,good,yes
old,medium,employed,good,yes
old,low,unemployed,poor,no
old,low,self-employed,average,no
middle,low,unemployed,poor,no
young,medium,employed,average,yes
young,low,unemployed,poor,no
old,medium,self-employed,good,yes
middle,medium,employed,average,yes
middle,high,self-employed,good,yes
old,medium,unemployed,poor,no
young,medium,self-employed,average,yes
middle,low,employed,poor,no



4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (10 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 84, 76, 35, 47, 64, 95, 66, 89, 36, 84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 50, 51, 56, 84, 60, 59, 70, 63, 66, 50

(i) Find which class had scored higher mean, median and range.

(ii) Plot above in boxplot and give the inferences

